

# C1GALT1C1 (Cosmc) — Golgi membrane (GO:0000139) function-assignment review

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**Gene:** C1GALT1C1 (Cosmc), Homo sapiens, UniProt Q96EU7 **Seed hypothesis:** C1GALT1C1 has Golgi membrane (GO:0000139) **Evidence type / reference under review:** TAS, Reactome:R-HSA-1964505 (co-ref

P 37216524

**Focus:** function-assignment — does the gene product directly have this CC term?

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## Executive Judgment

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**Verdict:** Refuted / over-annotated (for the direct-localization claim).

C1GALT1C1 encodes **Cosmc**, the C1GALT1-specific molecular chaperone. Multiple independent primary studies establish that **Cosmc is an endoplasmic-reticulum (ER)-resident single-pass type II membrane protein**, and one study maps a specific **ER-retention determinant to its transmembrane domain**. It is the *client* enzyme it chaperones — **T-synthase / C1GALT1** — that resides in the Golgi, not Cosmc itself.

The GO:0000139 (Golgi membrane) annotation is **TAS from Reactome:R-HSA-1964505**, a pathway model of Golgi-lumenal O-glycosylation. It is best explained as a **compartment carry-over**: Reactome assigns the Golgi compartment to the glycosylation reaction and its participants, and Cosmc — a regulator/client-folding factor mentioned in that pathway context — inherited the Golgi-membrane location. UniProt's own subcellular-location statement is deliberately conservative ("Membrane; Single-pass type II membrane protein") and does **not** assert Golgi.

**Most important caveat:** No paper directly demonstrates Golgi residence of Cosmc, so the annotation is not experimentally grounded; conversely, direct ER localization is well supported. The safest curation action is to **replace GO:0000139 with the ER membrane term (GO:0005789)**. A minor caveat is that UniProt itself has not (yet) attached an ER-membrane GO CC term to Q96EU7, so a curator should add it from the primary literature.

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## Evidence Matrix

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| Citation                                                                  | Evidence type                             | Supports/<br>Refutes/<br>Qualifies | Claim<br>tested                   | Key finding                                                                                                                                           | Context                         | C<br>li                             |
|---------------------------------------------------------------------------|-------------------------------------------|------------------------------------|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|-------------------------------------|
| <p><a href="#">P 18695044</a></p> <p>(Ju et al., J Cell Biol 2008)</p>    | Localization + interaction (direct assay) | <b>Refutes</b> Golgi; supports ER  | Where does Cosmc reside?          | "Cosmc is an endoplasmic reticulum (ER)-localized ... chaperone that binds directly to human T-synthase"; T-synthase "resides in the Golgi apparatus" | Human cells; T-synthase folding | H<br>Fe<br>p<br>se<br>C<br>fr<br>(C |
| <p><a href="#">P 21262965</a></p> <p>(Sun et al., J Biol Chem 2011)</p>   | Structural/ mutagenesis localization      | <b>Refutes</b> Golgi; supports ER  | Determinant of Cosmc localization | 18-aa TMD "is essential for ER localization and confers ER retention"; TMD Cys required for dimer/ER retention                                        | Chimera & mutant studies        | H<br>M<br>E<br>m<br>st<br>re        |
| <p><a href="#">P 19923218</a></p> <p>(Aryal et al., J Biol Chem 2010)</p> | Direct in-vitro chaperone assay           | <b>Refutes</b> Golgi; supports ER  | Cosmc identity/ location          | "an endoplasmic reticulum-localized molecular chaperone termed Cosmc"                                                                                 | In vitro folding of T-synthase  | H<br>In<br>re<br>or<br>lo           |
| <p><a href="#">P 12122208</a></p> <p>(Ju &amp; Cummings, PNAS 2002)</p>   | Discovery/ functional                     | Qualifies (function, not location) | Cosmc function                    | Cosmc is the unique chaperone required for T-synthase activity                                                                                        | Human/ mammalian                | H<br>fu<br>es<br>ch<br>ro           |

| Citation                                                                             | Evidence type                    | Supports/<br>Refutes/<br>Qualifies | Claim<br>tested               | Key finding                                                                                                                                                                                                                                                                                                                | Context        | C<br>li                                       |
|--------------------------------------------------------------------------------------|----------------------------------|------------------------------------|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-----------------------------------------------|
| <div><div><div>P</div><div>37216524</div></div><div>(Erger et al., 2023)</div></div> | Mutant phenotype (human disease) | Qualifies (chaperonopathy)         | Cosmc as chaperone in disease | Germline C1GALT1C1 defect = congenital disorder of glycosylation via reduced T-synthase activity                                                                                                                                                                                                                           | Human patients | H<br>ch<br>ro<br>lo<br>st<br>re<br>th<br>an   |
| UniProt Q96EU7 (database)                                                            | Database/topology                | Qualifies                          | Topology & existing CC terms  | SL comment is an <b>inference (ECO:0000305)</b> : only "Membrane; Single-pass type II membrane protein" (cyto 1–6, TM 7–26, luminal 27–318). CC keywords list only "Membrane" — <b>no ER and no Golgi keyword</b> ; function hedged as "Probable chaperone". GO CC = GO:0000139 (TAS:Reactome), GO:0070062 (exosome (HDA)) | Curated record | M<br>U<br>ag<br>de<br>as<br>G<br>th<br>T<br>q |

| Citation                              | Evidence type                              | Supports/<br>Refutes/<br>Qualifies              | Claim<br>tested                   | Key finding                                                                                                                                                                                                                                                                                                                        | Context                                             | C<br>li                              |
|---------------------------------------|--------------------------------------------|-------------------------------------------------|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|--------------------------------------|
| Human<br>Protein Atlas<br>(C1GALT1C1) | Localization<br>(antibody IF,<br>database) | <b>Refutes/<br/>Qualifies</b> Golgi             | Independent<br>IF<br>localization | Subcellular<br>main location =<br><b>"Vesicles"</b> —<br>not Golgi                                                                                                                                                                                                                                                                 | Human cell<br>lines,<br>antibody-<br>based          | M<br>A<br>v<br>p<br>n<br>G<br>a<br>E |
| Reactome:R-<br>HSA-1964505            | Pathway/<br>database                       | <b>Competing /<br/>source of<br/>annotation</b> | Basis of<br>GO:0000139            | Reaction =<br>"C1GALT1<br>transfers<br>Galactose to the<br>Tn antigen<br>forming Core 1<br>glycoproteins";<br>compartment =<br><b>Golgi<br/>membrane,<br/>Golgi lumen.</b><br>This is a<br><b>C1GALT1 (T-<br/>synthase)<br/>catalytic<br/>reaction;</b><br>Cosmc inherits<br>the Golgi<br>compartment<br>by pathway<br>association | Reactome<br>ContentService<br>(fetched this<br>run) | L<br>C<br>l<br>e<br>c<br>c<br>c      |
| Reactome<br>fetch (this<br>run)       | Computational/<br>database                 | <b>Refutes direct<br/>claim</b>                 | Origin of<br>GO:0000139           | The Golgi term<br>originates from<br>the Golgi-<br>resident client's<br>reaction, not<br>from Cosmc                                                                                                                                                                                                                                | Programmatic<br>API                                 | H<br>c<br>p<br>c<br>c<br>l           |

| Citation | Evidence type | Supports/<br>Refutes/<br>Qualifies | Claim<br>tested | Key finding          | Context | C<br>li |
|----------|---------------|------------------------------------|-----------------|----------------------|---------|---------|
|          |               |                                    |                 | localization<br>data |         |         |

## GO Decision Table (leads — require curator verification)

| GO_ID      | Label                          | Aspect | Evidence basis                                                                                              | Assessment                                               | Curation lead                  |
|------------|--------------------------------|--------|-------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|--------------------------------|
| GO:0000139 | Golgi membrane                 | CC     | Reactome TAS (R-HSA-1964505 = <b>C1GALT1 Golgi reaction</b> )                                               | Pathway-compartment carry-over; no direct Cosmc evidence | <b>REMOVE or mark non-core</b> |
| GO:0005789 | endoplasmic reticulum membrane | CC     | Primary lit<br><div>P 18695044</div> ,<br><div>P 19923218</div><br><b>21262965</b> (TMD ER-retention motif) | Best-supported direct localization                       | <b>ADD (IDA)</b>               |
| GO:0070062 | extracellular exosome          | CC     | UniProt HDA proteomics                                                                                      | Secondary/high-throughput                                | Retain as non-core             |
| GO:0051082 | unfolded protein binding       | MF     | Chaperone for T-synthase folding ( <div>P 19923218</div> )                                                  | Direct chaperone activity                                | Consider ADD                   |
| GO:0006457 | protein folding                | BP     | Chaperone-assisted folding of C1GALT1                                                                       | Core process                                             | Consider ADD                   |

(Also saved as `go_decision_table.csv`.)

## GO Curation Implications

- **Term class:** Cellular Component.
- **Lead (requires curator verification): Do not retain GO:0000139 (Golgi membrane) as a direct localization for Cosmc.** Replace with **GO:0005789 (endoplasmic reticulum membrane)** — consistent with type II single-pass topology and the mapped TMD ER-retention motif.
- **Rationale:** The Golgi term is TAS carried from a Reactome pathway whose compartment reflects the glycosylation reaction (Golgi lumen) and the client enzyme C1GALT1/T-synthase, **not** Cosmc's own residence. Multiple primary papers place Cosmc in the ER.
- **If retention is preferred by policy** (because Reactome TAS terms are not auto-removed): at minimum **add GO:0005789 with experimental support (e.g., IDA from**

P 21262965

P 18695044

and flag GO:0000139 as non-core / pathway-context. GO:0070062 (extracellular exosome, HDA) is likely a proteomics secondary observation and non-core.

- Avoid defaulting to "protein binding"; the informative, better-supported term is ER membrane (CC) plus the established MF/BP of chaperone activity for T-synthase folding (e.g., GO:0051082 unfolded protein binding / GO:0006486 context via its client).

## Mechanistic Scope

- **Direct molecular function:** Cosmc is an ER-luminal-facing type II membrane chaperone that binds nascent/partly denatured **T-synthase (C1GALT1)** and promotes its folding, preventing aggregation and ubiquitin-mediated degradation

P 18695044

P 19923218

- **Direct location:** ER membrane, retained via its TMD

P 21262965

- **Downstream / pathway consequence (NOT Cosmc's own location):** Properly folded T-synthase traffics to and functions in the **Golgi**, synthesizing the core 1 O-glycan (T-antigen). Golgi is where the *product* enzyme acts — this is the effect that Reactome's Golgi compartment captures, and the likely origin of the Golgi mis-assignment to Cosmc.
- **Disease manifestation (downstream):** Loss/mutation of Cosmc → Tn syndrome / congenital disorder of glycosylation

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 — a phenotype of impaired chaperone function, not evidence of Golgi residence.

## Conflicts and Alternatives

- **Compartment carry-over (most likely):** Reactome models the O-glycosylation reaction in the Golgi; Cosmc inherits "Golgi membrane" by pathway association.
- **Client/chaperone conflation:** T-synthase (Golgi) vs Cosmc (ER) are routinely discussed together; the Golgi location of the client can be mis-attributed to the chaperone.
- **Exosome annotation (GO:0070062):** high-throughput proteomics; not a functional localization and does not support Golgi.
- **Paralog note:** A paralog **C1GALT1C1L (Q6BAM2)** exists but is a distinct gene; the seed accession Q96EU7 is canonical Cosmc, so the Golgi term is not a paralog mixup — it is a same-gene pathway-compartment carry-over.
- **Reactome confirmation (this run):** R-HSA-1964505 is explicitly the *C1GALT1* Golgi galactosyltransferase reaction (compartment: Golgi membrane/lumen), confirming the Golgi CC term reflects the client enzyme's reaction, not Cosmc.
- **No isoform/organism conflict found:** ER localization is reported for human Cosmc; the mechanism (TMD ER-retention) is intrinsic to the protein sequence.
- **No primary paper places Cosmc in the Golgi** — the competing hypothesis (Golgi residence) lacks direct experimental support.

## Knowledge Gaps

1. **Absence of an ER-membrane GO CC term in UniProt for Q96EU7.** Checked UniProt JSON: only Golgi (TAS) and exosome (HDA) CC terms are present. Matters because a curator replacing the Golgi term needs a positive ER term with proper evidence codes. Resolution: attach GO:0005789 with IDA citing

P 21262965

P 18695044

2. **Whether any Cosmc pool cycles to the ERGIC/cis-Golgi.** No evidence found for this; the TMD retention data argue against significant Golgi steady-state pools. Resolution: colocalization microscopy (Cosmc vs ER markers/calnexin vs Golgi markers/GM130) — already effectively addressed by prior ER-retention work.
3. **Policy question:** whether GO/Reactome TAS compartment terms should be auto-propagated to regulatory participants. Matters broadly for chaperones of Golgi enzymes. Resolution: curator/consortium policy decision.

## Discriminating Tests

- **Confocal colocalization / IF:** endogenous or tagged Cosmc vs ER (calnexin, PDI) and Golgi (GM130, TGN46) markers — expect ER, not Golgi (confirmatory).
- **Glycosidase sensitivity:** Endo H sensitivity of Cosmc N-glycans (if any) → ER-type high-mannose vs Golgi-processed complex glycans distinguishes compartment.
- **TMD swap experiments (already reported,**

P 21262965

replacing Cosmc TMD with a non-ER TMD relocalizes it — direct causal test of ER retention.

- **Proximity labeling (BioID/APEX) from an ER-anchored vs Golgi-anchored bait** to define Cosmc's steady-state neighborhood.

## Curation Leads (require curator verification)

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- **Action change:** Replace **GO:0000139 (Golgi membrane, TAS:Reactome)** → **GO:0005789 (endoplasmic reticulum membrane)** as the direct CC; or, if retaining, downgrade GO:0000139 to non-core/pathway-context and add GO:0005789.
  - **Candidate references + exact snippets to verify:**
    - **P 18695044**  
— "Cosmc is an endoplasmic reticulum (ER)-localized adenosine triphosphate binding chaperone that binds directly to human T-synthase"; "Formation of active T-synthase, which resides in the Golgi apparatus, requires a unique molecular chaperone, Cosmc".
    - **P 21262965**  
— "the 18 amino acid transmembrane domain (TMD) of Cosmc is essential for ER localization and confers ER retention to select chimeras".
    - **P 19923218**  
— "an endoplasmic reticulum-localized molecular chaperone termed Cosmc".
  - **Candidate replacement/added GO terms:** CC GO:0005789 (ER membrane); supporting MF GO:0051082 (unfolded protein binding); BP GO:0006457 (protein folding) / regulation of O-glycan biosynthesis via client.
  - **Suggested curator questions:** (1) Is Reactome TAS compartment appropriate as a direct CC for a regulator whose client lives in that compartment? (2) Should GO:0070062 (exosome, HDA) be treated as non-core?
  - **Suggested experiments:** ER/Golgi colocalization; Endo H assay; TMD-swap relocation (confirmatory of published result).
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## Provenance

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- UniProt Q96EU7 programmatic fetch (execute\_code, requests) returned: SL comment "Membrane; Single-pass type II membrane protein" (evidence ECO:0000305, i.e., curator inference); CC keyword only "Membrane" (no ER, no Golgi); function "Probable chaperone"; topology cyto 1–6 / TM 7–26 / lumenal 27–318; GO CC = GO:0000139 (TAS:Reactome), GO:0070062 (HDA).

- Human Protein Atlas API (search\_download) for C1GALT1C1 returned subcellular main location "Vesicles" (antibody-based IF) — not Golgi.
- QuickGO/EBI endpoint was unreachable during this run (connection timeout); the full GO annotation set could not be independently enumerated beyond UniProt's.
- Reactome ContentService fetch of R-HSA-1964505 (this run) returned displayName "C1GALT1 transfers Galactose to the Tn antigen forming Core 1 glycoproteins (T antigens)", schemaClass "Reaction", compartment ['Golgi membrane','Golgi lumen'].
- GO decision table saved as `go_decision_table.csv`.
- PubMed retrievals:

P 18695044

P 21262965

P 19923218

P 37216524

*Independent-resource cross-check (Iteration 2): neither UniProt (agnostic "Membrane", inferred) nor HPA (vesicular) supports Golgi residence; only Reactome TAS does. This reinforces the over-annotation conclusion.*

*All localization conclusions are drawn from primary literature and the UniProt record; the Golgi-inheritance interpretation is inference from the Reactome compartment model and is labeled as such.*